

Operations

Week 3 – Live Class

Gerardo Berbeglia
Associate Professor
Melbourne Business School



PROCESS FLOW DIAGRAM:

Components



Storage areas or queues



Tasks or Operations



Flow of material or customers

THREE MEASURES OF PROCESS PERFORMANCE

Cycle Time (time/unit)

- Average time between the output of successive units (e.g. 5 min/unit)
- **Flow rate or throughput rate** is the inverse of the cycle time (e.g. $1/5$ units/min = 0.2 units/min)

Flow time or manufacturing lead time or service lead time

- The average amount of time it takes a unit to get through the process (includes waiting time).

Inventory or Work in process (WIP)

- Average number of units in the process at any point in time (includes the units waiting to be processed).

Process Utilisation and Resource utilisation

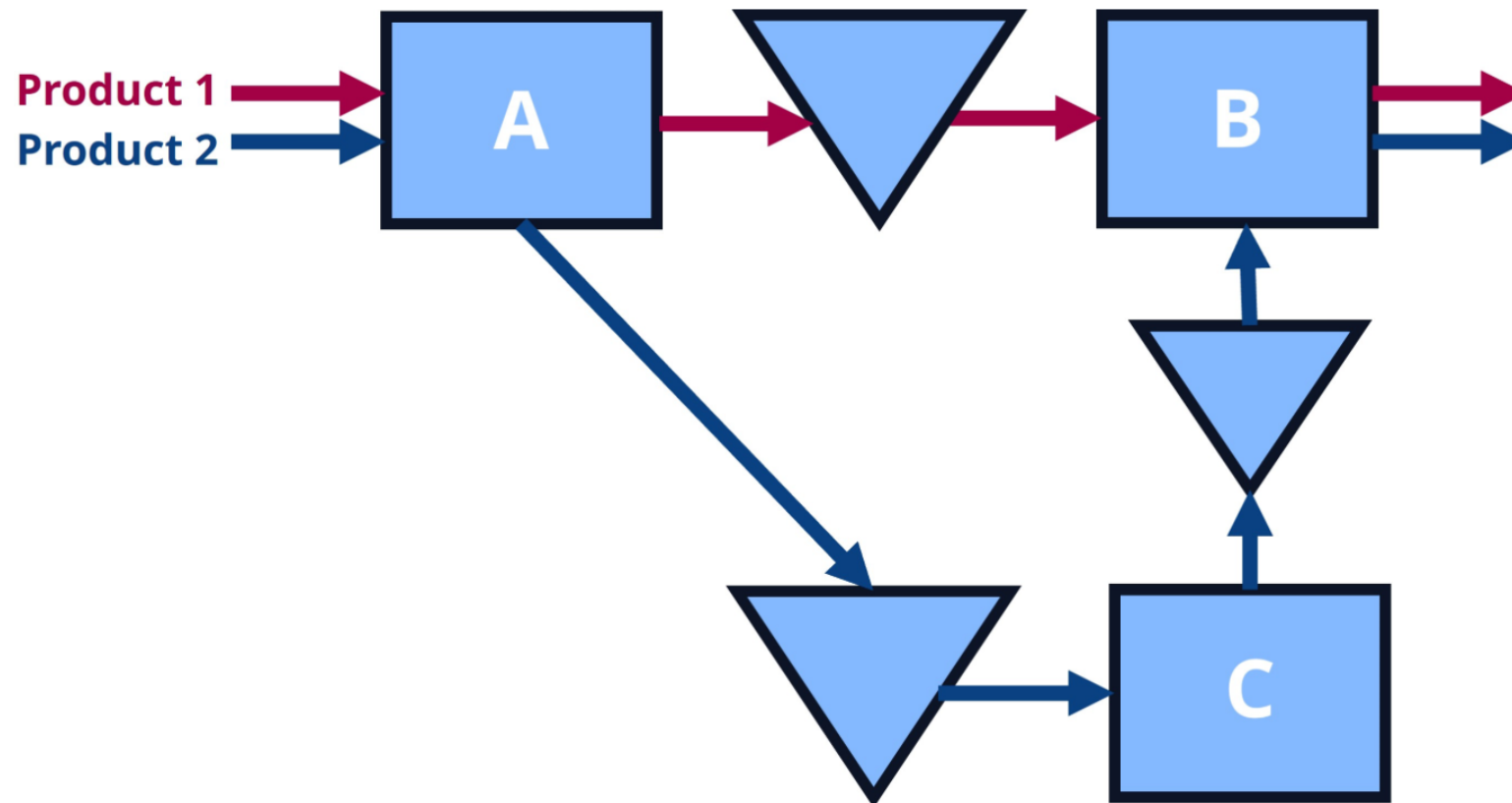
Process utilisation captures how much the process actually produce in comparison to how much it can produce if it were running at full speed.

$$\text{Process utilization} = \frac{\text{Flow rate}}{\text{Process capacity}}$$

The utilisation of a resource is

$$\text{Utilization of a resource} = \frac{\text{Flow rate}}{\text{Capacity of resource}}$$

Processes with different type of units



Implied utilisation

Utilisation: It carries information about excess capacity but not about excess demand.

Implied utilisation: captures the mismatch between the capacity requested from a resource by demand and the capacity currently available at the resource.

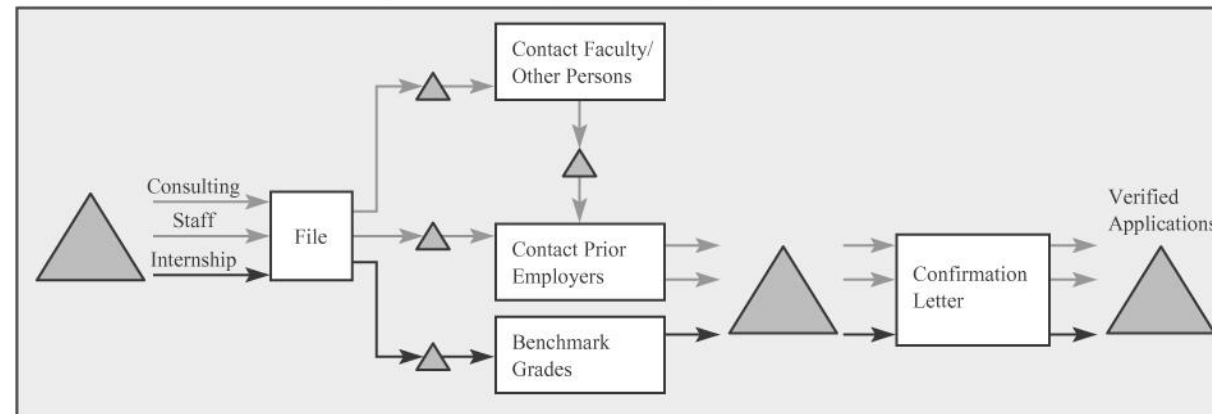
$$\text{Implied utilization} = \frac{\text{capacity requested by demand}}{\text{Capacity of resource}}$$

What does it mean if a resource has an implied utilisation of above 100%?

The bottleneck is the resource with the highest implied utilisation.

Dealing with Multiple types of flow units

FIGURE 3.10 Process Flow Diagram with Multiple Product Types



An employment verification agency receives resumes from consulting firms and law firms with the request to validate information provided by their job candidates.

Note that the product mix complicates the process analysis.

How to determine the bottleneck and capacity in a multi-product situation?

TABLE 3.5 Finding the Bottleneck in the Multiproduct Case

	Activity Time	Number of Workers	Available Capacity	Requested Capacity [Applications/Hour] Workload				Implied Utilization
				Consulting	Staff	Interns	Total	
File	3 [min./appl.]	1	1/3 [appl./min.] = 20 [appl./hour]	3	11	4	18	18/20 = 90%
Contact persons	20 [min./appl.]	2	2/20 [appl./min.] = 6 [appl./hour]	3	0	0	3	3/6 = 50%
Contact employers	15 [min./appl.]	3	3/15 [appl./min.] = 12 [appl./hour]	3	11	0	14	14/12 = 117%
Grade/school analysis	8 [min./appl.]	2	2/8 [appl./min.] = 15 [appl./hour]	0	0	4	4	4/15 = 27%
Confirmation letter	2 [min./appl.]	1	1/2 [appl./min.] = 30 [appl./hour]	3	11	4	18	18/30 = 60%

Choose the unit of flow: an application (we do not differentiate between application types)

We calculate the **available capacity** (app/per hour) for each resource.

We calculate the **requested demand** (app/hour) for each resource.

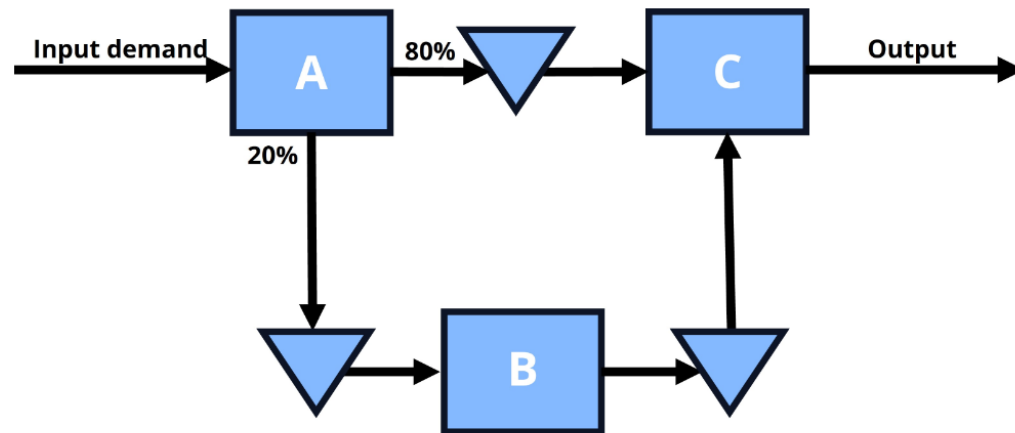
Bottleneck: Resource with **highest** implied utilisation

Consider a process that has three steps: A, B, and, C.

There are two types of request from customers.

Type 1: They need to pass by Steps A and C only. They represent 80% of the requests.

Type 2: They need to pass by Steps A, then Step B and finally Step C. They represent 20% of the requests.



The total number of requests at busy times is six requests per hour.

Step A: It takes five minutes to handle each request. There is only one employee.

Step B: It takes 29 minutes to handle each request. There is only one employee.

Step C: It takes eight minutes to handle each request. There is only one employee.

Discussion in tables: where is the bottleneck?

A



B



C

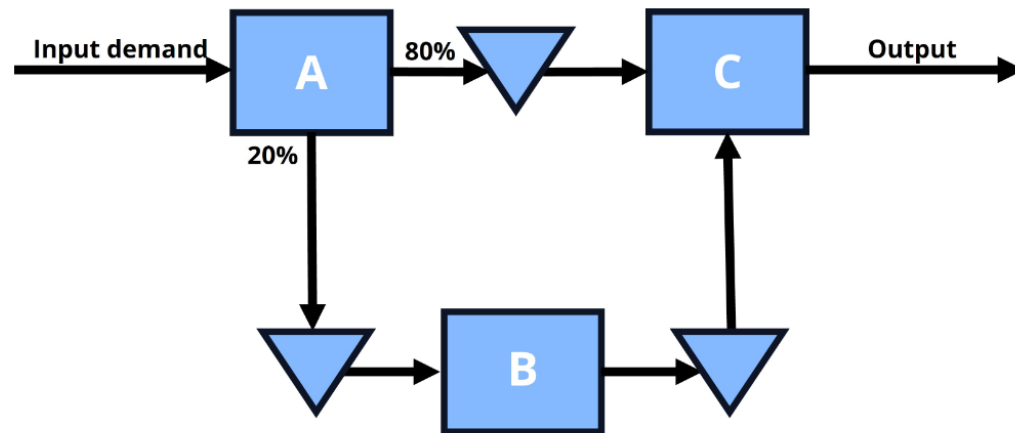


Consider a process that has three steps: A, B, and, C.

There are two types of request from customers.

Type 1: They need to pass by Steps A and C only. They represent 80% of the requests.

Type 2: They need to pass by Steps A, then Step B and finally Step C. They represent 20% of the requests.



The total number of requests at busy times is six requests per hour.

Step A: It takes five minutes to handle each request. There is only one employee.

Step B: It takes 29 minutes to handle each request. There is only one employee.

Step C: It takes eight minutes to handle each request. There is only one employee.

We first calculate the capacity requested by the demand at each step:

Step A: Six requests per hour

Step B: $6 \times 0.2 = 1.2$ requests per hour

Step C: Six requests per hour.

Now we calculate the capacity of each resource:

Step A: $\frac{60}{5} = 12$ requests per hour

Step B: $\frac{60}{29} \simeq 2.068$ requests per hour

Step C: $\frac{60}{8} = 7.5$ requests per hour.

Finally, we calculate the implied utilisation at each step:

Step A: $\frac{6}{12} = 0.5$ or 50%

Step B: $\frac{1.2}{2.068} \simeq 0.58$ or 58%

Step C: $\frac{6}{7.5} = 0.8$ or 80%

The bottleneck is step C and the implied utilisation is 80%.

Discussion: Implied utilisation

- How could implied utilisation serve as a useful tool at your company or within an industry you are familiar with?
- What are the main challenges associated with calculating it?
- What would be the “optimal” level of implied utilisation for that firm or industry?

JOB SHOP: Characteristics

Multiple paths

Independent work stations buffered each other by WIP inventory

High WIP inventories

Production scheduling method: human judgment

General purpose machines

Labor: not finely divided

Opportunities for competitive advantage

Human resources

Rapid adaptation to satisfy customers needs – fast product development

FLOW SHOPS: Characteristics

- Small number of paths
- Tasks: high coordination
- WIP inventories: low
- High utilization rates
- Capacity is simple to know
- Small variability
- Labor: finely divided

Opportunities for competitive advantage

- Constant reduction in costs
- Better service / delivery

Examples: automobile factory, laptop factory

Design issues in product Layout: Line Balancing

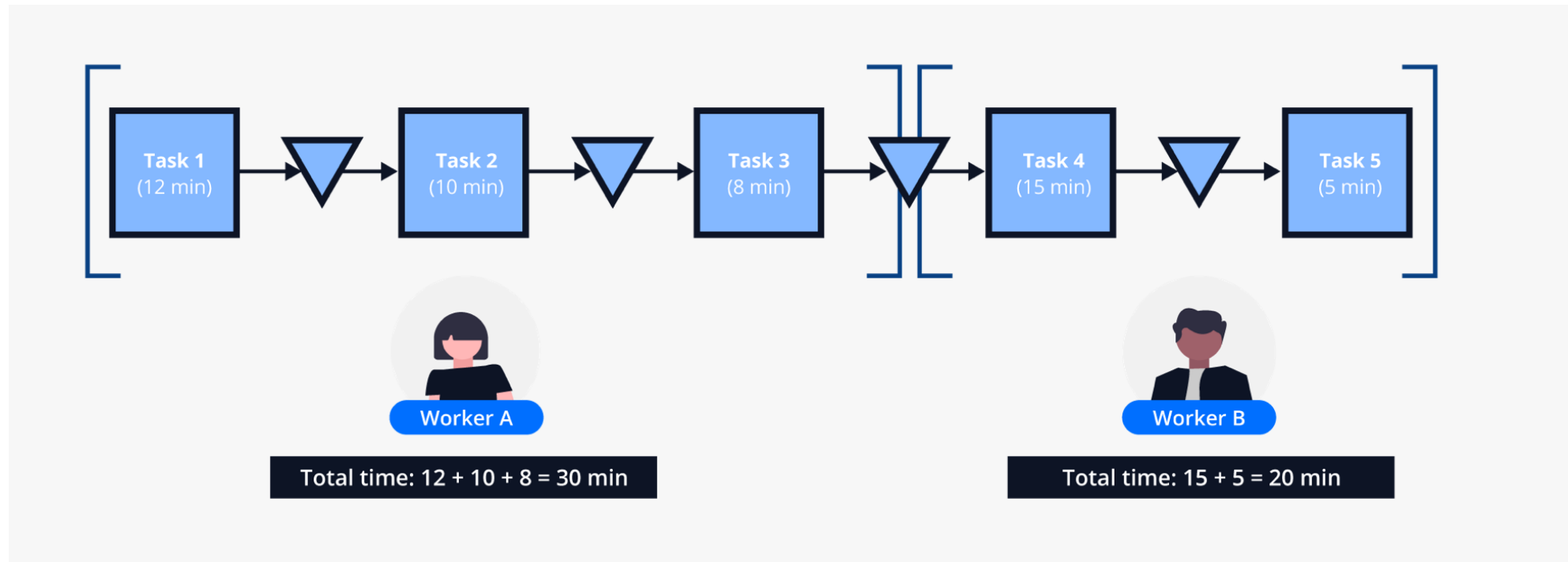


The challenge

- Single product
- Small work units called *tasks* (indivisible)
- Each task i takes a certain time T_i
- The product moves through a conveyor belt
- Work is done at workstations
- How should we assign tasks to workstations?
- Objective: maximize process capacity (minimise cycle time)
- Other possible objective: minimise number of workstations

Line Balancing

Example



Labor content and direct Labor cost



Labor content per unit: total working time for the production of 1 unit.

- Example of 3 workers: 13min/unit + 11 min/unit + 8min/unit = 32 min per unit.

How do we know **the cost of direct labor** (per unit)?

- We need:
 - The flow rate (#units / hr)
 - Total amount of wages per unit of time (\$/hr)
 - **Cost of direct labor:** total wages / flow rate (\$/units)

Average Labor utilisation



We want to capture the percentage of the time that it is spent in producing output units.

Total time per unit = labor content per unit + total idle time (per unit)

Average labor utilisation =
labor content (per unit) / total time per unit

Power Toys Inc

Power Toys inc. produces a small remote-controlled toy truck on a conveyor belt with nine stations. Each station has, under the current process layout, one worker assigned to it. Stations and processing times are summarized in the following table.

Station	Task	Processing time in seconds
1	Mount battery units	75
2	Insert remote control receiver	85
3	Insert chip	90
4	Mount front axle	65
5	Mount back axle	70
6	Install electric motor	55
7	Connect motor to battery unit	80
8	Connect motor to rear axle	65
9	Mount plastic body	80

Part A

a) What is the bottleneck in the process?

Station	Task	Activity Time (in Sec.)	Capacity (Units/hour)	Idle Time (in Sec.)	Utilization
1	Mount battery units	75	48.00	15	83.33%
2	Insert remote control receiver	85	42.35	5	94.44%
3	Insert chip	90	40.00	0	100.00%
4	Mount front axle	65	55.38	25	72.22%
5	Mount back axle	70	51.43	20	77.78%
6	Install electric motor	55	65.45	35	61.11%
7	Connect motor to battery unit	80	45.00	10	88.89%
8	Connect motor to rear axle	65	55.38	25	72.22%
9	Mount plastic body	80	45.00	10	88.89%

The bottleneck in the process is Station 3 (Insert Chip process) with a capacity of 0.67 units per minute [40 units per hour].

Part B

If one person works in each work station, what is the capacity, in toy trucks per hour, of the assembly line?

The capacity of the process is the capacity of the bottleneck
= 0.67 units per minute or 40 units per hour.

Part C)

What is the direct labour cost for the toy truck with the current process if each worker receives \$15/hour, expressed in dollars per toy truck?

Direct labor cost per unit (assuming that the flow rate is capacity-constrained and not demand-constrained):

$$= 9 * \$15 / 40 = \$3.38 / \text{unit}$$

Part D

What would be the direct labour cost for the toy truck if work would be organized in a work cell, that is, one worker performed all tasks?
(Assume that the activity times would remain unchanged)

If the process operated as a work cell, each worker would perform all 9 tasks, and the total labor content for each worker would be the total activity time = sum of task times = 665 seconds

Capacity = $60 \times 60 / 665 = 5.41$ trucks/hour/cell

The total system capacity with 9 worker cell is = $9 \times 5.41 = 48.72$ trucks/hour

Direct labor per unit in work cell layout = $9 \times \$15 / 48.72 = \2.77

Part E



What is the utilisation of the worker in station 2?

$$= 85/90 = 94.4\%$$

Staff reduction



Because of a drastically reduced forecast, the plant management has decided to cut staffing from nine to six workers per shift. Assume that (i) the nine tasks in the above table cannot be divided; (ii) the nine tasks are assigned to the six workers in the most efficient way possible; and (iii) if one worker is in charge of two tasks, the tasks have to be adjacent (i.e. one worker cannot work on tasks 1 and 3).

Part f

How would you assign the nine tasks to the six workers?

Line Balancing with 6 Workers

Worker	Task	Activity Time
1	Mount battery units	75
2	Insert remote control receiver	85
3	Insert chip	90
4	Mount front axle	65
	Mount back axle	70
	Total	135
5	Install electric motor	55
	Connect motor to battery unit	80
	Total	135
6	Connect motor to rear axle	65
	Mount plastic body	80
	Total	145

Part G

What is the new capacity of the line (in toy trucks per hour)?

Worker	Total Activity Time	Capacity (Units/hr.)
1	75	48.00
2	85	42.35
3	90	40.00
4	135	26.67
5	135	26.67
6	145	24.83

Capacity of the line = 24.83 trucks/hour

AIC Netbooks



What was the planned output per line per shift?

Note that the total productive time available in a shift is 8 hours = 28,800 seconds.

Also note that the planned cycle time is determined by the maximum station time which equals 41 seconds for station 9.

Hence, the planned output per line per shift = $28,800/41 = 702$ units.

Planned output

4 lines, 2 shifts/day, 6 days/week, 52 weeks/year
Each line has a planned output of 702 units/shift

Planned output

$$= 4 \times 2 \times 6 \times 52 \times 702 = 1,752,192 \text{ units/year}$$

$$= 1,752,192 / 12 = 146,016 \text{ units/month}$$

Actual output achieved

Note that achieved cycle time for each line = 50 seconds
(stations 2 and 6 are slow compared to their designed time)

For a shift , the output = $\frac{(60 \times 60 \times 8)}{50} = 576$ units /shift/line

The yearly output for one shift from each line = $576 \times 6 \times 52 = 179,712$ units/shift/line

With four lines operating for 2 shifts per day,

The actual output for the plant will be

= $179,712 \times 4 \times 2 = 1,437,696$ units/year

or **119,808 units/month**

Assessment of theoretical efficiency

Sum of task times = 396 seconds (labor content per unit)

The longest step is station 9 which implies a planned cycle time is 41 seconds.

The number of assembly line workers for the planned operation is 10.

Planned total time per unit

=

Theoretical efficiency (average labor utilisation) = $396 / 410 = 96.6\%$

$$10 \text{ employees} \times 41 \text{ sec/ unit} = 410 \text{ sec/ unit}$$

Assessment of actual efficiency

The assembly line currently requires 12 employees

The actual station time (labor content per unit) = 443 seconds

The total time per unit = 12 employees x 50 seconds/unit = 600 seconds

The actual efficiency achieved (labor utilisation) = $443/600 = 73.8\%$

COGS breakup

Processor and other integrated circuit boards	\$50.40	22.83%
SSD+ Ram	\$67.20	30.43%
Other Components	\$76.80	34.78%
Labor (Assembly, Packing, Loading, Delivery)	\$4.80	2.17%
Manufacturing Overhead	\$21.60	9.78%
Manufacturer COGS	\$220.80	100.00%

Note that 88% of COGS is in components. Direct labour and Manufacturing overhead accounts for 12%. Labour is a very small component of overall COGS.

Possible Recommendations

Redesign the assembly line operation to lower the cycle time.

Redesign the product such as simplifying the time it takes for assembly. This can help lower the cycle time.

Additional training

Provide more incentives for continual improvement.

Utilisation Problem

The process is capacity constrained and the entire process produces 36 units in one hour. It is a **line process** that has 5 steps.

U1 through U5 denote the utilization at steps 1 through 5 respectively.

Currently, there is a single worker at each step and the utilizations are as follows:

$U1 = 4/30$, $U2 = 4/15$, $U3 = 4/5$, $U4 = 1$, $U5 = 2/5$.

Answer the following questions based on the above data and information.

- What is the capacity of step 5?
- Which step is the bottleneck?
- Which process step has the highest capacity?

Utilisation Problem

a) The process is capacity constrained and the entire process produces 36 units in one hour.

$$U_1 = 4/30, U_2 = 4/15, U_3 = 4/5, U_4 = 1, U_5 = 2/5.$$

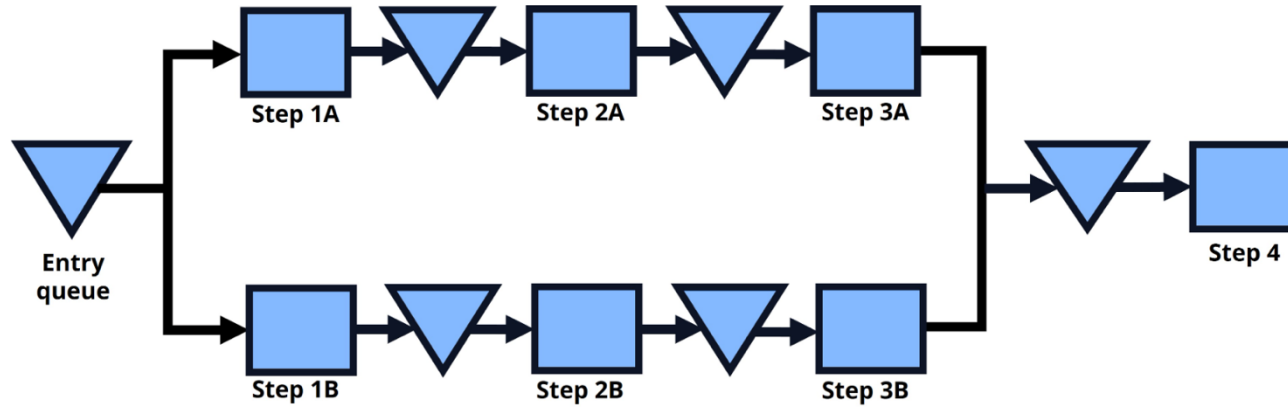
$$\begin{aligned}\text{Utilisation of Step 5} &= \text{Flow rate} / \text{Cap of step 5} = 2/5 \\ &= 36/\text{hr} / \text{Cap of step 5} = 2/5\end{aligned}$$

$$\text{So, Cap of step 5} = 36/\text{hr} / (2/5) = 90 \text{ units per hour}$$

b) The bottleneck is the step with the lowest capacity. In this problem, this is the step with the highest utilisation. This is step 4.

c) In this problem, the step with the highest capacity is the step that has the lowest utilisation. This is step 1.

Consider the following process flow diagram.



Half of the output units go through path 1 and the other half go through path 2, where:

Path 1: Step 1A ->Step 2A ->Step 3A ->Step 4

Path 2: Step 1B ->Step 2B ->Step 3B ->Step 4

Suppose each of the following six steps takes one minute per unit: Step 1B, Step 2B, Step 3B, Step 1A, Step 2A and Step 3A.

Select the most accurate statement to finish the sentence below.

Step 4 is the (only) bottleneck if:

- ☐ It takes more than 30 seconds per unit.
- ☐ It takes more than 1 minute per unit.
- ☐ Its capacity is less than 60 units per minute.
- ☐ It takes more than 2 minutes per unit.

Exercise 1

Explanation

If step 4 did not exist, the overall system capacity would be 2 units per minute. This is because each of the six steps takes one minute, meaning that with two parallel lines, the maximum flow rate would be 1 unit per minute per line.

Now, we need to determine whether step 4 acts as the bottleneck. A step is a bottleneck if it limits the overall system capacity. Since we established that without step 4, the system can process 2 units per minute, step 4 would be the bottleneck if its capacity is lower than 2 units per minute. In other words, step 4 would restrict the system if its processing time exceeds 30 seconds per unit.

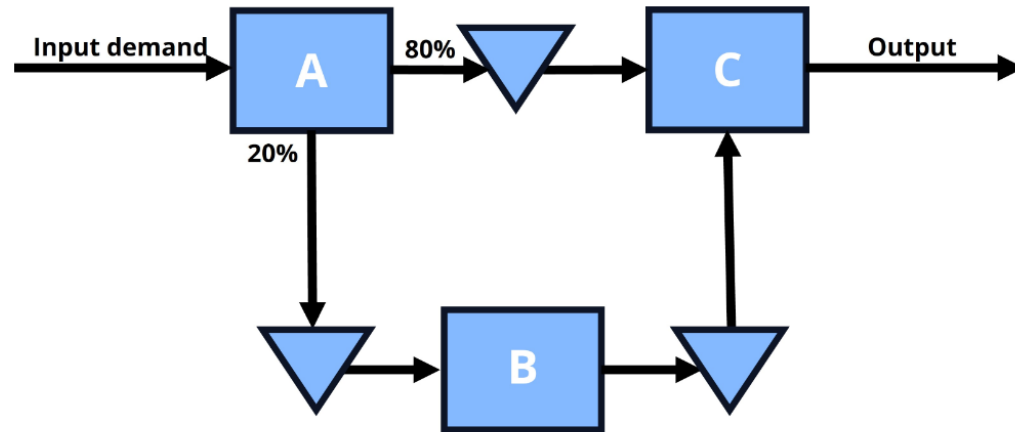
Exercise 1

Consider a process that has three steps: A, B, and, C.

There are two types of request from customers.

Type 1: They need to pass by Steps A and C only. They represent 80% of the requests.

Type 2: They need to pass by Steps A, then Step B and finally Step C. They represent 20% of the requests.



The total number of requests at busy times is six requests per hour.

Step A: It takes five minutes to handle each request. There is only one employee.

Step B: It takes 29 minutes to handle each request. There is only one employee.

Step C: It takes eight minutes to handle each request. There is only one employee.

Exercise 2:

What step is the bottleneck?

Explanation

We first calculate the capacity requested by the demand at each step:

Step A: Six requests per hour

Step B: $6 \times 0.2 = 1.2$ requests per hour

Step C: Six requests per hour.

Now we calculate the capacity of each resource:

Step A: $\frac{60}{5} = 12$ requests per hour

Step B: $\frac{60}{29} \simeq 2.068$ requests per hour

Step C: $\frac{60}{8} = 7.5$ requests per hour.

Finally, we calculate the implied utilisation at each step:

Step A: $\frac{6}{12} = 0.5$ or 50%

Step B: $\frac{1.2}{2.068} \simeq 0.58$ or 58%

Step C: $\frac{6}{7.5} = 0.8$ or 80%

The bottleneck is step C and the implied utilisation is 80%.

Exercise 2: What step is the bottleneck?

Key Takeaways



New method to find the bottleneck under more complex processes: We need to calculate the implied utilisation.

Broad classification of processes and their characteristics: job shop, batch job, flow shops

Measurement of the process capacity in line process.

New method to increase capacity: Line balancing

New process performance measures and their relationships: idle time, labor content, labor utilisation, and direct labor cost.

Case study highlights the financial benefits of having a balanced line and achieving the planned capacity.

Connect with us



@MelbBSchool



/MBScarleton



/MelbourneBusinessSchool



@MelbourneBSchool



Melbourne-business-school